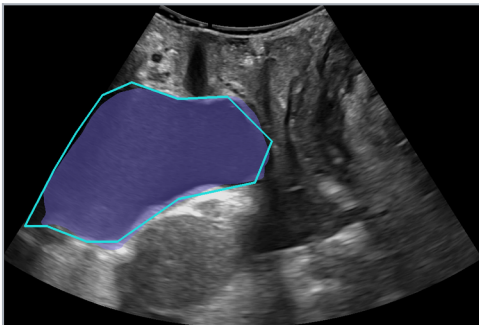


1 Segmentation



U-Net lumen mask on the B-mode bladder frame, 256×256

mask $\rightarrow$ centroid $\hat{C}$
image $\rightarrow$ contrast, centering,
boundary sharpness

2 Quality judgment

Q — weighted geometric mean
of image and mask sub-scores

attained range 0.64
 $\rightarrow$ 47 distinguishable states

92.4% of the range is read
off the ultrasound image

*decides whether the frame can
instruct, and what it falls
short of*

3 Corrective action

$\hat{e} = A - \hat{C}$ lateral, in-plane

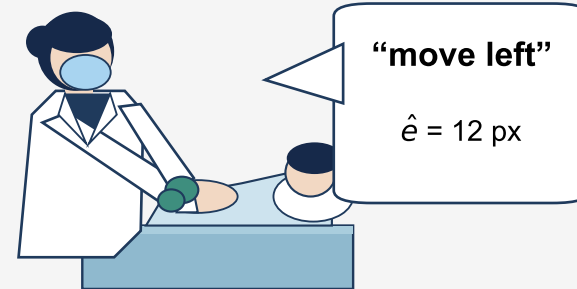
contrast $\leq 0 \rightarrow$ check bladder filling
 $|\hat{e}| < 1.64\sigma = 8.1 \text{ px} \rightarrow$ hold
otherwise $\rightarrow$ move left / move right

the word names the direction;
 $\hat{e}$ travels beside it in pixels,
unrounded

both boundaries derived, not tuned
 $\sigma = 4.92 \text{ px}$ centroid noise

Human operator

language, displayed or spoken



word + $\hat{e}$

Robotic probe holder

action token, translated by lookup



(move left, $\hat{e}$)
 $\rightarrow v_x = -k \hat{e}$, saturated,
other axes zero

hold $\rightarrow$ zero twist
check filling $\rightarrow$ gate

execution: §4.1, design only